

Claude Code Quick Reference

AIA Copilot — Claude Code Training

What Is Claude Code?

Claude Code is an **agentic coding tool** that runs in your terminal. Unlike autocomplete tools (Copilot) or AI editors (Cursor), Claude Code operates as an autonomous engineer — it reads your codebase, plans changes, writes code, runs commands, and fixes its own errors.

You provide intent. Claude provides implementation.

Installation

```
curl -fsSL https://claude.ai/install.sh | bash
```

Verify: `claude --version`

Requirements: Node.js 18+, Git, Anthropic API key or Claude Pro/Max subscription

Essential Commands

Command	What It Does
<code>claude</code>	Start Claude Code in current directory
<code>/clear</code>	Reset conversation (files stay on disk)
<code>/compact</code>	Compress conversation history (save tokens)
<code>/cost</code>	Show token usage and cost
<code>/model</code>	Switch models (opus, sonnet)
<code>/help</code>	Show all available commands
<code>/doctor</code>	Run diagnostics
<code>/memory</code>	Show what Claude remembers
<code>/permissions</code>	View/edit permission rules
<code>Escape</code>	Cancel current operation

Context Control (@ Mentions)

Syntax	What It Loads
<code>@filename.js</code>	Single file
<code>@src/</code>	Entire folder
<code>@src/routes/*.js</code>	Pattern match
<code>@https://docs.example.com</code>	Web page content

Why it matters: Every file Claude reads costs tokens. @ mentions load exactly what you need — nothing more.

CLAUDE.md — Your AI Policy File

Create `CLAUDE.md` in your project root. Claude reads it automatically before every task.

```
# Project Conventions

## Code Style
- Use const/let, never var
- async/await over callbacks
- Guard clauses for validation

## Architecture
- Business logic out of route handlers
- Database access in dedicated modules

## Git
- Conventional commits: feat:, fix:, docs:
```

Think of it as: ESLint + Architecture Guide + Team Playbook — for an AI developer.

Scope	File	Shared?
Personal	<code>~/.claude/CLAUDE.md</code>	No — your machine only
Personal (project)	<code>CLAUDE.local.md</code>	No — gitignored
Team/Org	<code>CLAUDE.md</code> in repo	Yes — everyone who clones
Subdirectory	<code>subfolder/CLAUDE.md</code>	Yes — loaded in that folder

The Agentic Loop

```
YOUR INTENT → READ → PLAN → EXECUTE → VERIFY
                ↑       |
                └───┬───┘
                Self-healing loop
```

Claude reads errors, diagnoses the cause, fixes the code, and retries — automatically.

Prompting Patterns

Bad (vague)

```
Fix the bug.
```

Good (precise)

```
The POST /api/tasks endpoint returns 500.  
Check @src/routes/tasks.js for the error.
```

Spec-Driven (best)

```
# Feature: Dashboard Notifications  
## What: Toast notifications when metrics change  
## Why: Users miss important changes  
## Constraints: No external libraries, auto-dismiss 5s  
## Done when: Notifications appear and auto-dismiss
```

Key Workflows

Plan Before Building

```
I want to add rate limiting. Analyze the codebase first  
and tell me your approach before making any changes.
```

Architectural Review

```
Identify the 3 biggest architectural weaknesses.  
Rank them by production risk.
```

Recovery

Metrics functionality disappeared after a bad merge.
Reconstruct based on @src/routes/tasks.js patterns
and CLAUDE.md conventions.

Security Audit

Review the entire codebase and fix anything that
violates our security rules in CLAUDE.md.

Git Workflow

Create branch feature/export, build the feature,
commit with conventional message, merge to main.

Context Management

Tool	When to Use
<code>/clear</code>	Start fresh (conversation resets, files stay)
<code>/compact</code>	Running low on context, compress history
<code>.claudeignore</code>	Exclude files from auto-loading (like .gitignore)
<code>@ mentions</code>	Load specific files instead of everything

Context window $\approx$ 200K tokens. Treat it as a budget.

Pricing (as of March 2026)

Plan	Cost	Best For
Pro	\$20/mo	Individual developers
Max 5x	\$100/mo	Power users (5x usage)
Max 20x	\$200/mo	Teams and heavy usage
API	Pay per token	Automation and CI/CD

Troubleshooting

Problem	Fix
Claude stops mid-task	<code>exit</code> → <code>claude</code> → "Continue where you left off"
Port in use	Tell Claude: "Switch to port 3456"
Permission errors	Check <code>/permissions</code>
Slow responses	Try <code>/model sonnet</code> for faster output
Context too large	<code>/compact</code> or <code>/clear</code>
File not found	Use <code>@</code> with the exact path
<code>tmpclaude-*</code> files	Safe to ignore — auto-cleanup

The 7 Skills of AI-Assisted Development

1. **Problem Framing** — Define intent and constraints precisely
2. **Context Engineering** — Design CLAUDE.md and manage token budget

3. **Agent Delegation** — Coordinate parallel AI workflows (Day 3)
 4. **Critical Judgment** — Evaluate output for risk and quality
 5. **Verification Design** — Build quality gates (security audit, policy enforcement)
 6. **Toolchain Architecture** — Connect tools via MCP (Day 3)
 7. **Intelligence Orchestration** — Shape human-AI collaboration
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Resources

- **Anthropic Docs:** <https://docs.anthropic.com/en/docs/claude-code>
 - **Claude Code GitHub:** <https://github.com/anthropics/claude-code>
 - **Prompt Engineering:** <https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering>
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